

# Response to Reviewers

A Controlled Reassessment of Support Vector Regression on the California Housing Benchmark with Feature Scaling, Feature Selection and Hyperparameter Tuning

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Dear Dr Addula and Mr Mohite,

Thank you for the decision of major revision, and my thanks to the three reviewers for reports that were detailed, technically specific and, in two important places, correct about errors I had made. The revision that follows is substantial: the manuscript has been rewritten around a new set of experiments, and two of the earlier version’s central claims are withdrawn rather than defended.

Three changes deserve to be stated at the outset, because they alter the paper’s central argument rather than merely elaborating it.

## 1. The reference value was misread, and the correction removes the paper’s central inconsistency.

Reviewer 1 (point 2) and Reviewer 3 both observed that an unscaled baseline of  $R^2 = -0.054$  cannot explain a previously reported  $R^2$  of 0.60, and that the earlier manuscript never accounted for the gap. They were right, and on re-reading Preethi et al. closely I found the cause: 0.60 **is not their SVR result**. It is the score of their linear, ridge and polynomial-ridge models. Their SVR-RBF model attains  $R^2 \approx 0.15$  with  $MSE \approx 1.12$ , read from their Figs. 7 and 8 (their paper reports no numerical table). The earlier version of my manuscript attributed the linear family’s score to their SVR. That was my error. With the correct value in place, the configuration they describe reproduces at  $R^2 = -0.020$  with library defaults and 0.514 when  $C$  and  $\epsilon$  are tuned — values that bracket 0.15 — and no unexplained gap remains. This is Section 1.2 and Section 5.1 of the revised manuscript.

## 2. The proposed feature-specific scaling strategy does not work, and is now reported as a negative result.

Reproducing the prior study’s uniform scaling directly, as the Editor asked in point 1, required me to compare it against my own feature-specific assignment on an identical split. Uniform standardisation gives  $R^2 = 0.7453$ ; the hand-assigned feature-specific strategy gives 0.6925 on the same split with the same features. The strategy I presented in the earlier version as the preprocessing advance is therefore *worse* than plain standardisation. It is now reported as such, and the workflow’s scaling stage is described as “any reasonable scaling” rather than as a contribution.

## 3. The attribution of 95.7% of the gain to scaling was an artefact of ablation order.

Evaluating the full factorial design rather than a single path shows that the apparent contribution of scaling ranges over 0.7441 in  $R^2$  depending only on where it sits in the sequence. The single-path figure reported earlier sat near the top of that range. The order-independent Shapley value is 0.2723. Scaling remains the largest contributor, but the earlier percentage is withdrawn.

I have also added a disclosure of my use of a large language model, as Reviewer 2 requested and the Editor endorsed in point 12.

Every point raised by the Editor and all three reviewers is answered below, with the location of each change. Reviewer comments are set in *grey italic*. All numerical values quoted in this letter are generated from the same result records that generate the manuscript’s tables and figures, so the two documents cannot disagree.

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## 1. Response to the Editor’s twelve points

### E1. Directly reproduce baseline performance.

*The uniform scaling SVR configuration used in previous work should be reproduced directly and its performance compared to the authors’ feature-wise scaling method.*

**Response.** Done, and it produced a negative result that I report. A new experiment (Section 5.1, Tables 2 and 3) re-runs the configuration described by Preethi et al. over 5 seeds: eight raw features, RBF kernel, a grid over  $C$  and  $\varepsilon$  only,  $\gamma$  at the library default, 80/20 partition. It yields  $R^2 = -0.020$  untuned and 0.514 tuned, bracketing their reported 0.15. Table 5 then compares uniform standardisation against my feature-specific assignment on one identical split across all feature sets and training sizes: uniform gives 0.7453, feature-specific gives 0.6925. The feature-specific strategy loses. This is stated in the abstract, in Section 5.1, and again in Section 6.2 among the withdrawn claims.

### E2. Fair model comparison.

*SVR and baseline algorithms should be trained / evaluated with similar quantities of data, rather than trained-set sizes that differ by an order of magnitude or more.*

**Response.** Corrected. Section 5.2 (Table 6) retrains all 10 models with identical features, identical training rows and an identical search budget of 20 randomised iterations with three-fold cross-validation, reported at both 3,000 rows and the full training set. The tuned SVR attains 0.7501 (rank 4) at 3,000 rows and 0.7702 (rank 4) on the full set. Section 5.8 (Table 16) additionally shows that the exact full-data fit takes only ?? s, so the subset restriction had no computational justification and has been removed from the main results rather than defended. A Nyström approximation is reported for completeness but is not needed at this scale.

### E3. Statistical significance testing.

*The confidence interval equation is incorrect. Proper paired statistical tests should be performed to compare models.*

**Response.** Both parts corrected. The earlier version reported  $\bar{r} \pm 1.96s$ , which is a normal-approximation prediction interval for a single fold, not a confidence interval for the mean. Section 5.3 (Table 7) now reports the interval three ways so the difference is explicit: the correct  $\bar{r} \pm t_{0.975, J-1} s / \sqrt{J}$  gives [0.769, 0.783], the Nadeau–Bengio interval gives [0.759, 0.792], and the earlier expression gives [0.756, 0.795]. The ten-fold interval of the earlier version moves from [0.621, 0.765] to [0.667, 0.719]. The argument that the earlier interval did not overlap a competing value is withdrawn. For model comparison, all models are now evaluated on 10 *identical* splits and compared with the Nadeau–Bengio corrected resampled  $t$ -test, since resampled training sets overlap and the ordinary paired  $t$  is anti-conservative; the uncorrected  $t$  and the Wilcoxon signed-rank test are reported alongside it (Table 8).

### E4. Ablation study should be improved.

*The impact of each stage of the proposed pipeline should be tested more rigorously, including the order in which they are applied or alternative methods to several pipeline stages.*

**Response.** Rebuilt. The ablation is now a four-factor full factorial design — scaling, derived features, selection, tuning — and *every one* of its 32 cells is evaluated (Table 10). Each component receives a Shapley value, which averages over all 24 orderings and is the unique attribution that is both order-independent and additive. The result changed the paper’s conclusion: scaling appears to contribute between  $-0.0239$  and  $+0.7202$  depending on placement, and its order-independent value is  $0.2723$ , not the 95.7% previously claimed. Alternative methods at each stage are also swept: three scaling strategies, four kernels each tuned over its own hyperparameters (Table 14), and grid versus randomised search at three budgets (Table 15).

#### **E5. Validate predictive inference.**

*Consider spatial validation of the model, or at minimum holdout feature selection within each cross-validation fold to avoid overly optimistic estimates of model performance.*

**Response.** Both were done, and this produced the most consequential result in the paper. Section 5.5 (Table 11) crosses two partitioning schemes with two selection schemes. Random ten-fold cross-validation gives  $0.7496$ ; cross-validation over ten spatially contiguous blocks, obtained by  $k$ -means on standardised centroids, gives  $0.2316 \pm 0.7525$ . The optimism of random partitioning,  $0.5180$ , is larger than every pipeline effect measured anywhere in this study. Feature selection repeated inside each fold changes the estimate by  $0.0000$ ; the effect is small because the selected sets are stable across folds, but it is reported because the earlier version could not have known that. Fully nested estimates, with both selection and hyperparameter search inside each outer fold, are  $0.7532$  (random) and  $-0.5344$  (spatial). Section 6.3 now states plainly that results of this kind measure interpolation within a sampled region, not prediction for an unsampled one.

#### **E6. Limit claims to dataset and configuration studied.**

*Given that only one benchmark dataset was used, claims in the discussion should be limited to those pertaining to the California Housing dataset and specific configuration tested.*

**Response.** Done throughout. A new subsection, *Scope of the claims* (Section 1.5), states that all quantitative claims are restricted to the datasets, feature sets, search spaces and implementations described in Section 4, and that the necessity of scaling for distance-based kernels is long established and not claimed here as a discovery. The title now reads “A Controlled Reassessment ...on the California Housing Benchmark”. Section 6.2 is devoted to what the evidence does *not* support. The abstract closes on the same restriction. Every general statement about SVR has been removed or bounded.

#### **E7. Define proposed workflow.**

*The overall processing pipeline should be explicitly defined. It may also be helpful to include a diagram that illustrates the different stages of the pipeline and what data is input and output to each stage.*

**Response.** A new Section 3, *Proposed workflow*, defines the pipeline as nine stages (S1–S9) with stated inputs and outputs, *before* any data are analysed. Figure 1 is a new diagram showing each stage and the data flowing between them, with the stages fitted on the training fold enclosed in a dashed boundary; Table 1 states the same information tabularly. The scaler assignment, previously a hand-made table specific to one dataset, is now stated as an algorithm

(Algorithm 1) that assigns a scaler to each feature from training-fold statistics alone, so it can be audited and applied elsewhere.

**E8. Definition of methods is lacking.**

*Much more detail is needed in the description of how models are trained, how hyperparameters were optimized, what cross-validation strategy was used, how features were engineered and models compared, and how the ablation study was performed.*

**Response.** Section 3 now specifies each of these separately: Section 3.1 the SVR formulation with every symbol defined; Sections 3.2–3.4 feature derivation, ensemble importance and selection; Section 3.5 the scaler rule as pseudocode; Section 3.6 the search spaces and budgets; Section 3.7 the evaluation protocol, which distinguishes explicitly between the inner cross-validation used for hyperparameter selection and the outer resampling used for performance estimation — the ambiguity Reviewer 2 identified; and Section 3.7 the ablation protocol. Section 4.3, *What is held fixed*, states for each comparison exactly which factor varies and which are held constant.

**E9. Claim of novelty should be improved.**

*Currently the paper suggests that the authors have broadly “rediscovered” SVR. The authors should avoid making this claim unless a new novel contribution is demonstrated.*

**Response.** Withdrawn entirely. Section 1.4 now opens: “The contribution of this paper is methodological and diagnostic rather than algorithmic. No new learning algorithm is proposed.” Section 2.1 places the work explicitly within the existing scaling literature (Sola and Sevilla; Hsu et al.; Singh and Singh; Fernández-Delgado et al.) and states that the paper does not claim that result. What is claimed is narrower and, I hope, defensible: a corrected reading of one published result, a direct reproduction of it, an order-independent ablation procedure, a corrected evaluation protocol, and two negative results.

**E10. Manuscript language and formatting requires revision.**

*Figure / table captions, references, math notation, numbering of results all require revision. In addition, the overall flow of the manuscript can be improved by revising section headings.*

**Response.** The manuscript was rewritten rather than patched. Specifically: figure titles embedded in images have been removed so the caption is the sole description; all tables are referenced by arabic numeral matching their captions, replacing the Roman numerals of the earlier version; every reference has been rechecked and DOIs added; all mathematical symbols are defined where introduced (Section 3.1) and collected in a notation table (Table 2); section headings are now declarative statements of content, and Section 3 (workflow) now precedes Section 4 (experimental setup) as Reviewer 2 requested. On numerical consistency I took a structural approach rather than a proofreading one: **no number is typed by hand anywhere in this manuscript**. Every experiment writes a JSON record; a generator script turns those records into every table, every figure and every in-text value. The class of error Reviewer 1 identified in point 9 is therefore now impossible rather than merely corrected.

**E11. Use additional datasets.**

*Test your proposed workflow on additional datasets. Reviewer #2 particularly felt that this would improve the authors’ argument that their method does not only work well for California Housing.*

**Response.** Section 5.6 (Table 12) applies the workflow unchanged — same scaler rule, same selection rule, same search space, no dataset-specific adjustment — to King County house sales (21,613 transactions, 18 numerical predictors) and Ames Housing (1,460 sales, 36 usable numerical predictors). The result is not the clean confirmation I expected, and I report it as it came out.

Under plain uniform standardisation the ordering  $\text{scaling} > \text{tuning} > \text{selection}$  is reproduced on all 3 datasets: scaling contributes +0.1869, +0.2710 and +0.1211, tuning +0.0730, +0.1155 and +0.0283, and selection +0.0157, +0.0053 and +0.0068.

Under the distribution-aware scaler rule that this revision introduced, it is reproduced on only 1 of 3. On King County the rule makes scaling the smallest contributor (+0.0805, against +0.2202 for tuning); on Ames its Shapley value is negative (−0.1558), and applying it first costs 0.6455 in  $R^2$ . Ames is the smallest and widest of the three (1022 training rows, 36 candidate features), which is where per-feature scaler assignment estimated from training statistics is least reliable.

I have therefore withdrawn the rule as a contribution rather than presenting the additional datasets as a confirmation. Section 5.6 reports both strategies side by side, Figure 6 plots both, and Section 6.2 lists the rule among the claims the evidence does not support. This is the third of the paper’s negative results, and it is the one that most directly answers Reviewer 2’s concern: testing the workflow on other datasets did not corroborate it, it falsified part of it. I also state the limit of the evidence in Section 6.3: all three are residential-property datasets, so this tests transfer across datasets within a domain, not across domains.

## E12. AI-use statement.

*Reviewer #2 found large portions of the text to be generated by AI. Authors should consider adding a statement disclosing use of any LLM.*

**Response.** Added as a declaration subsection, *Use of generative artificial intelligence*. It discloses that I used a large language model (Anthropic Claude) as an assistive tool for drafting and copy-editing prose, for implementing and refactoring the experiment scripts, and for generating tables and figures from the result records; and it states that the research questions, experimental design, interpretation and all scientific claims are my own, that I have reviewed and verified every number, statement and reference, and that no text appears that I have not read and checked against the underlying results. Separately, the specific symptoms Reviewer 2 identified — an excessive number of thin subsections, and mismatches between figures and captions — have been fixed by consolidating the subsections and regenerating every figure with its caption from the same source.

## 2. Response to the internal editorial feedback

### I1. Declaration section with individual subheadings.

**Response.** The Declarations section now carries *Ethical approval*, *Consent to participate* and *Consent to publish* as separate subheadings, each answered individually. All three are “Not applicable”, with the reason stated: the study analyses publicly available aggregated datasets and involves no human participants, animal subjects, or identifiable personal data.

### I2. Data availability statement in the manuscript above the References.

**Response.** Added above the References, with direct links: the California Housing dataset via `sklearn.datasets.fetch_california_housing` and the StatLib archive; King County via OpenML <https://www.openml.org/d/42731>; Ames via OpenML <https://www.openml.org/d/42731>.

[org/d/42165](https://doi.org/10.26434/chemrxiv-2023-42165). The analysis code, the JSON records of every experiment reported, and the scripts that generate every table and figure are at <https://github.com/emmanueladutwum123/svr-california-housing>.

### I3. Dual publication declaration.

**Response.** Added. An earlier version was deposited as a preprint on arXiv (arXiv:2605.08660); no part of the work is published in or under consideration at any other peer-reviewed journal. The declaration states that the present manuscript is a substantially revised version of that preprint, that the reading of the reference result is corrected, and that the experimental content of Sections 5.1–5.8 is new.

## 3. Response to Reviewer 1

I am grateful for this report. Points 2 and 9 identified a genuine error in the earlier manuscript, and correcting it changed the paper.

### R1.1 Reframe as a controlled reassessment, not a new general finding.

*The improvement of SVR performance through feature scaling is a long-established result ... The manuscript should therefore be reframed not as presenting a new general finding about SVR, but as a controlled reassessment of the specific result reported by Preethi et al. (2025).*

**Response.** Accepted in full; this is now the frame of the paper, beginning with its title. Section 1.5 states that scaling for distance-based kernels is long established and cites Sola and Sevilla (1997), Hsu et al. (2003) and Singh and Singh (2020) for it; Section 2.1 positions the work within that literature; Section 6.2 repeats that the paper establishes nothing about SVR in general.

### R1.2 The unscaled baseline cannot explain the previously reported 0.60.

**Response.** You were right, and finding out why changed the paper. The 0.60 figure is the score of their *linear family*; their SVR is  $R^2 \approx 0.15$ . My earlier manuscript misattributed it. Section 5.1 now reproduces their configuration directly, under the same 80/20 split over 5 seeds, obtaining  $-0.020$  untuned and  $0.514$  tuned, which bracket their value; and Table 5 gives the direct comparison of their uniform scaling ( $0.7453$ ) against my feature-specific strategy ( $0.6925$ ) on an identical split. Thank you for pressing on this.

### R1.3 SVR trained on 3,000 rows against competitors trained on 14,448.

**Response.** Corrected; see E2 above. All 10 models now receive identical rows, features and search budget (Table 6), reported at both sizes. Table 16 shows the full-data fit costs ?? s, so the subset had no justification. Nyström approximation is reported but proves unnecessary at this scale.

### R1.4 Paired tests on identical splits rather than a bare confidence interval.

**Response.** Adopted, using both references you gave. Section 5.3 evaluates every model on 10 identical splits and reports paired differences with Nadeau–Bengio (2003) corrected intervals and  $p$ -values, with the uncorrected  $t$  and Wilcoxon alongside for reference (Dietterich, 1998). Against XGBoost the difference is  $-0.0728$  with corrected  $p = 1.9 \times 10^{-7}$ , where the uncorrected test would have given  $p = 1.2 \times 10^{-10}$  — I report both precisely because the discrepancy makes your point.

### R1.5 The ablation follows only one sequence.



**Response.** Corrected; see E4. All 32 cells of the four-factor design are evaluated and attribution is by Shapley value over all 24 orderings. The order-dependence you anticipated is large: scaling’s apparent contribution spans 0.7441 in  $R^2$ . The manuscript now states explicitly that single-path increments should be read as upper bounds for whichever component was applied first.

#### **R1.6 Spatially blocked validation.**

**Response.** Added, following Roberts et al. (2017); see E5. Spatial blocking costs 0.5180 in  $R^2$  relative to random partitioning — more than any pipeline effect in the paper. Figure 5 shows the blocks and the resulting estimates.

#### **R1.7 Justify the engineered features, the 40/30/30 weights and the top-12 selection.**

*In particular, Coastal Proximity is based only on latitude difference, while Location Score is defined through the product of latitude and longitude ... preferably through sensitivity analysis.*

**Response.** Two changes. First, you were right that neither name was defensible, and Section 3.2 now says so in the text rather than leaving the labels to imply more than they measure. *Coastal proximity*,  $|\text{Latitude} - 34.05|$ , is stated to be a one-dimensional north–south displacement from a reference latitude near the Los Angeles coastline — explicitly *not* a distance to the coast, which would require the coastline geometry and both coordinates. *Location score*,  $(\text{Latitude} \times \text{Longitude})/1000$ , is stated to be an interaction term between the two coordinates with no geographic units, to be read as an unnamed coordinate interaction rather than a location measure. Both are retained as *candidate* features whose value is decided empirically at Stage S4 rather than by their names, and the ablation reports how little the derived block contributes once selection and tuning are present. Second, Section 5.7 (Table 13) sweeps every hand-chosen constant: across ?? weight combinations on the simplex test  $R^2$  varies only between ?? and ?? (range ??), and the smallest overlap between any swept selection and the default top-twelve is ?? of twelve; varying  $k$  from 4 to 18 moves  $R^2$  between ?? and ??; varying the scaler thresholds over nine combinations moves it by ??. The honest conclusion is that these constants are not load-bearing, which I take to be the appropriate justification for constants with no theoretical derivation.

#### **R1.8 Feature selection performed outside the cross-validation folds.**

**Response.** Corrected rather than merely acknowledged. Selection is now repeated inside every fold, and Table 11 reports selection-inside against selection-outside under both partitioning schemes; the difference is 0.0000. Fully nested estimates following Varma and Simon (2006) are 0.7532 (random) and  $-0.5344$  (spatial). The effect is small here because the selected sets are stable across folds — feature-selection stability is now recorded in the result files — but the earlier version had no way of knowing that, which was your point.

#### **R1.9 Findings rest on a single benchmark; restrict conclusions and moderate broad statements throughout the title, abstract, discussion and conclusion.**

*The findings are based on a single, extensively studied benchmark dataset. Unless additional datasets are included, all conclusions should be restricted to the California Housing dataset and the specific experimental configuration used. Broad statements regarding the general performance of SVR should be moderated throughout the title, abstract, discussion, and conclusion.*

**Response.** Addressed on both of the paths you offer, since I took the first one as well as the second. Additional datasets *are* now included: Section 5.6 applies the workflow unchanged to King County house sales and Ames Housing, and reports that the ordering of the components

is reproduced on both under uniform standardisation, while the distribution-aware scaler rule introduced in this revision fails to transfer and is withdrawn as a contribution. Independently of that, the claims have been moderated in each of the four places you name. The **title** is now “A Controlled Reassessment ...on the California Housing Benchmark”. The **abstract** closes on the sentence “Conclusions are restricted to the datasets and configurations examined here.” The **discussion** opens each of its supported claims with “Within the California Housing dataset and the configurations tested here”, and Section 6.2 states that the paper establishes nothing about SVR in general — only that one published comparison was configured in a way that disadvantaged one of the models it compared. The **conclusion** restates the restriction explicitly. A new Section 1.5 states the scope of every quantitative claim before any result is presented. I would also note the limit of the new evidence rather than overstate it: all three datasets are residential-property datasets, so Section 5.6 tests transfer within a domain, not across domains, and Section 6.3 says so.

#### **R1.10 Numerical and reporting inconsistencies throughout.**

**Response.** Every number in the manuscript has been recalculated from scratch, and the class of error has been eliminated structurally: each experiment writes a JSON record and a generator produces every table, figure and in-text value from those records, so no value is typed by hand and the text, tables and figures cannot diverge. Rounding is standardised to four decimal places for  $R^2$  and to the reported precision elsewhere. The monetary conversions you flagged have been removed rather than corrected, since with the target capped at \$500,001 they were more misleading than informative. Mean and median are now labelled distinctly wherever both appear. Figure and table numbering is generated automatically.

### **4. Response to Reviewer 2**

This report identified the two structural weaknesses I found hardest to answer — that the pipeline was never actually formalised, and that the manuscript showed signs of over-generated text. Both are addressed directly.

#### **R2.0 (general remark) Figures 3–6 are unnecessary for a well-known dataset, or at least not at their current size.**

**Response.** Agreed and acted on: the exploratory figures of the earlier version — the distribution and correlation displays that occupied Figures 3–6 — have been removed from the manuscript entirely rather than merely reduced. The dataset is, as you say, thoroughly documented elsewhere, and Section 4.1 now conveys the two facts that actually matter for this paper’s argument in a sentence of text: the feature ranges that motivate the scaling stage, and the census recording cap at \$500,001 that bounds achievable fit. Every figure that remains reports a result rather than describing the data.

### **Major issues**

#### **R2.1 The contribution is weak; the pipeline is explained vaguely and should be formalised with a diagram.**

**Response.** Accepted. Section 1.4 now states plainly that no new learning algorithm is proposed and that the contribution is methodological and diagnostic. The pipeline is formalised in a new Section 3 as stages S1–S9 with defined inputs and outputs, with Figure 1 as the diagram you suggested and Table 1 as its tabular statement, and the scaler assignment promoted from a hand-made table to an auditable algorithm (Algorithm 1). I would add that the revision now also offers a methodological contribution that the earlier version did not: the order-independent



Shapley attribution for preprocessing ablations (Section 3.7), which is what revealed that the earlier 95.7% figure was an artefact.

**R2.2 Why compare only against Preethi et al.? Why only four references in Section 2.2?**

**Response.** Section 2 has been expanded and restructured into three subsections — scaling in kernel methods, the California Housing benchmark, and model comparison and attribution — with the benchmarking literature discussed in terms of settings and reported performance rather than merely listed. Preethi et al. remain the specific object of the reassessment, because a controlled reproduction requires a specific configuration to reproduce, but they are no longer the only point of comparison: Section 5.2 places the tuned SVR against 10 models under equal conditions, and the finding that gradient-boosted trees lead on this benchmark is connected to the broader literature in Section 2.2.

**R2.3 Disclose LLM use.**

**Response.** Disclosed in full; see E12. I have also acted on the symptoms you identified rather than only on the disclosure: the thin subsections have been consolidated, and every figure is now regenerated from the result records together with its caption, which removes the figure/caption mismatches.

**R2.4 The training procedure is unclear — 3-fold HPO, then 10-fold evaluation on what data?**

**Response.** This was genuinely ambiguous in the earlier version, and I have separated the two loops explicitly. Section 3.7 now states that hyperparameter selection uses an inner  $k$ -fold cross-validation *on the training portion only*, and that performance is estimated by an outer resampling scheme over held-out data that the inner loop never sees; Section 3.7 states which is used for each reported quantity. Where both selection and hyperparameter search occur inside the outer folds, the estimate is labelled “nested” (Table 11).

**R2.5 Section 3 should follow Section 4 (previous version).**

**Response.** Done: the proposed workflow (now Section 3) precedes the experimental setup (now Section 4), so the proposal is stated before the data used to evaluate it.

**R2.6 What is the regression equation mentioned on Page 7?**

**Response.** That sentence referred to nothing defined in the manuscript and has been removed. The SVR decision function is now given once, in Section 3.1, with every symbol defined.

**R2.7 Figure titles duplicate captions; Figure 9’s title and caption disagree; figures referenced by title appear unreferenced.**

**Response.** All embedded figure titles have been removed, so the caption is the sole description. Every figure is regenerated from its result record, and each is referenced in the text by its  $\text{\LaTeX}$  label rather than by title, so the reference, the caption and the content are guaranteed to correspond.

**R2.8 Table captions use arabic numerals but the text uses Roman; format is not Springer Nature style.**

**Response.** Corrected: all tables are numbered and referenced with arabic numerals in Springer Nature style, using the `sn-jnl` class throughout.

**R2.9 References use an incorrect format; academic papers should have DOIs.**

**Response.** The bibliography has been rebuilt in `sn-mathphys-num` style, and every one of the 29 references now resolves to a persistent link. Twenty-one carry a DOI, including two that were missing and have been added (Vapnik’s monograph, 10.1007/978-1-4757-2440-0, and Shapley’s 1953 chapter, 10.1515/9781400881970-018). The remaining eight have no DOI because their venues do not register them — five are *Journal of Machine Learning Research* articles, two are NeurIPS proceedings papers, and one is a technical report — so each of those now carries the publisher’s stable URL instead, verified to resolve to the correct paper. No reference is now without either a DOI or a permanent link.

**R2.10 Section 4.1.2 is not self-contained — define  $n, \phi, b, \mathbf{w}, \gamma, \alpha, y, \mathbf{x}, \|\mathbf{w}\|^2$ .**

**Response.** Agreed — a paper explicitly about SVR should not omit its formulation. Section 3.1 now gives the primal objective, the  $\varepsilon$ -insensitive loss, the dual and the kernel expansion, with every symbol you list defined at first use, and Table 2 collects the full notation.

**R2.11 Table 3 placement; Section 4.2.2 should precede 4.2.1; results in Section 4 belong in Results.**

**Response.** Section 4 has been restructured as you describe. The derived features are now listed before the table that summarises them; the ordering of the two subsections is reversed; and every empirical value that appeared in the old Section 4 has been moved into Section 5, so Section 4 states the setup and reports no results.

**R2.12 Hyperparameter tuning: why discretise  $C$  and  $\varepsilon$  for a randomised search? Why not a grid? Why tune only the RBF kernel? Why only 20 iterations if the search takes 9 seconds? Table 7 is unnecessary.**

**Response.** Every part of this was a fair criticism and all are now addressed empirically rather than argued.  $C$ ,  $\varepsilon$  and  $\gamma$  are now drawn from continuous log-uniform distributions, so the randomised search is a genuine randomised search. Section 5.8 (Table 15) reports the exhaustive grid alongside randomised search at three budgets: the  $\gamma$ -fit grid reaches  $\gamma$  in  $\gamma$  s, against  $\gamma$  at 20 iterations and  $\gamma$  at 200. Since the search costs seconds, the larger budget is used throughout the revision and the small-budget limitation is removed rather than defended. Every kernel is now tuned over its own hyperparameters, not only the RBF (Table 14). The previous version’s Table 5 has been deleted and its content folded into the text.

**R2.13 Why is stratification needed in a regression problem?**

**Response.** It is not needed, and the manuscript now says so explicitly rather than implying otherwise. Section 3.6 states that stratification is not required in regression and that the deciles are used for one narrow purpose: to remove a source of variation between configurations that are being compared against each other, so that a difference between two cells reflects the configuration rather than a difference in the target distribution of their partitions. Nothing in the paper’s conclusions rests on it — the repeated-split analysis of Section 5.3, which is where every significance claim is made, draws fresh unstratified partitions, and the spatially blocked design of Section 5.5 is the deliberate alternative where dependence between observations genuinely matters. I have kept the device and justified it rather than removing it, since removing it would add noise to the ablation comparisons without changing any conclusion; I am happy to drop it entirely if you would prefer.

**R2.14 Were the other algorithms also tuned? Did they use the same pipeline?**

**Response.** They are now, and they do. Section 5.2 gives every model the same twelve features, the same training rows, the same scaling stage and the same search budget of 20 randomised iterations with three-fold cross-validation. Section 4.3 states this explicitly. In the earlier version the comparison models were used at their defaults, which — as you imply — made the ranking claim uninterpretable.

**R2.15 The ablation needs detail: did you start from Preethi et al.’s setting and add components progressively? Was anything held fixed?**

**Response.** The design is now stated in full in Section 3.7, and it no longer proceeds by a single progressive path. The baseline cell is the configuration described by Preethi et al. (no scaling, raw features, defaults), at  $R^2 = -0.0277$ ; from there all 32 cells of the four-factor design are evaluated, so no component’s contribution depends on what was held fixed while it was introduced. The best cell is uniform / derived / all / tuned at 0.7807.

**R2.16 Section 7.1 repeats earlier results; consider merging Sections 6 and 7.**

**Response.** The Discussion has been rewritten as three subsections that make claims rather than restate numbers — what the evidence supports, what it does not support, and threats to validity — with no repetition of the Results tables.

**R2.17 If the search takes 9 seconds, the stated limitations do not hold — why sample 3,000 instances, and why not increase the HPO budget? LIBSVM’s complexity is not a reason for a less ambitious setting.**

**Response.** You were right and I have not defended any of it. The 3,000-row subset is removed from the main results (Table 16 shows the full fit takes ?? s), the search budget is increased and justified empirically (Table 15), and the LIBSVM-complexity argument has been deleted from the limitations. What remains in Section 6.3 are limitations I can actually defend: the spatial optimism, the census cap at \$500,001, the 1990 vintage of the data, the within-domain nature of the additional datasets, and the reading error of roughly  $\pm 0.02$  in the values taken from Preethi et al.’s figures.

## Minor issues

**R2.18 Add a reference for the hedonic pricing model (Section 1.1).**

**Response.** Added: Rosen (1974), with Limsombunchai (2004), Fan et al. (2006) and Park and Bae (2015) for the machine-learning alternative.

**R2.19 Add references for the California Housing dataset.**

**Response.** Added: Pace and Barry (1997) for the dataset’s introduction and Pedregosa et al. (2011) for its scikit-learn distribution.

**R2.20 Section 1.4’s second sentence is confusing — 10 new features but top 12 used — and contradicts the experiments.**

**Response.** The confusion was real: ten features are *derived*, giving eighteen candidates alongside the eight raw features, from which twelve are selected. The earlier phrasing implied twelve were selected from ten. Section 1.4 and Section 3.2 now state the counts explicitly at each stage, and Figure 1 shows the dimensionality at each transition.

**R2.21 Section 2.3’s title should reflect its content.**

**Response.** Section 2 has been restructured into three subsections with titles that state their content: *Feature scaling in kernel methods*, *The California Housing benchmark*, and *Comparing models and attributing gains*.

**R2.22 Table 1 is unnecessary (previous version).**

**Response.** Deleted, with the small amount of information it carried folded into the text.

**R2.23 Section 3.2 repeats “characteristics” in one sentence.**

**Response.** Rewritten.

**R2.24 How can “wide residual spread” be observed in the caption of Figure 7?**

**Response.** It could not be, from that figure. The caption is rewritten to describe only what the panels show — residuals centred near zero with dispersion increasing at higher predicted values, and the vertical band of true values at 5.0 produced by the census recording cap.

## 5. Response to Reviewer 3

Thank you for the recommendation of minor revision and for a report that identified the same central problem as Reviewer 1 while also being precise about where the caveats belonged.

**R3.1 The central comparison does not hold together: Stage A yields  $-0.054$  against Preethi et al.’s  $\approx 0.60$ , and the gap is never explained.**

**Response.** This was the most important observation in the three reports, and pursuing it changed the paper. The gap could not be explained because the premise was wrong: 0.60 is the score of their linear family, and their SVR is  $\approx 0.15$ . Rather than either of the two options you offered, I have taken the third: the misreading is corrected and stated openly (Sections 1.2 and 5.1), their configuration is reproduced directly, and the reproduced values  $-0.020$  and  $0.514$  bracket their actual result. The “explains the prior result” language is now supported by a direct reproduction rather than by an assumed correspondence.

**R3.2 The training-set-size caveat must accompany the ranking claim wherever it appears, not only in Limitations.**

**Response.** Rendered unnecessary rather than relocated: all 10 models are now trained on identical data (Section 5.2), so the ranking no longer carries an asymmetry that needs a caveat. Both training sizes are reported, and the ranking is stated with its training size attached in the abstract, in Section 5.2 and in the conclusion.

**R3.3 The Introduction asserts “this does not restrict SVR as a learning algorithm” before any evidence is presented.**

**Response.** Removed and replaced by the question being tested. Section 1.3 now poses four research questions (Q1–Q4), each cross-referenced to the section that answers it, so the Introduction states what is asked rather than what will be concluded.

**R3.4 The claim that full-dataset training “would be expected to yield lower RMSE” is untested.**

**Response.** It is now tested rather than asserted. Table 16 fits the same tuned configuration at three training sizes; RMSE falls from ?? at 3,000 rows to ?? on the full training set, so the direction of the earlier claim is confirmed — but it is now a measurement.

### **R3.5 Soften “confirms” to “supports” or “is consistent with”.**

**Response.** Done throughout. “Confirms” now appears only where a specific prediction was tested directly and the outcome was as predicted, as in R3.4 above.

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I am grateful for the care evident in all three reports. The requirement to reproduce the prior configuration directly is what uncovered my misreading of the reference value, and the requirement to evaluate the full ablation design is what showed that my headline attribution was an artefact of ordering. The paper is more modest than it was and, I believe, considerably more solid.

Yours sincerely,

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